

Question

It is known that $[\text{XeF}][\text{AsF}_6]$ reacts with H_2O in anhydrous HF according to the following equilibrium:



When equimolar amounts of $[\text{XeF}][\text{AsF}_6]$ and H_2O react in anhydrous HF at $-78\text{ }^\circ\text{C}$, the adduct $[\text{H}_3\text{O}][\text{AsF}_6] \cdot 2\text{XeF}_2$ is initially formed. Subsequently, $[\text{XeF}][\text{AsF}_6]$ reacts with free XeF_2 in a 1:1 ratio to yield compound **A**. This adduct further reacts with free XeF_2 to form a 1:1 salt **B**. The anion of **B** is identical to that of **A**, while its cation contains oxygen with a coordination number of 2. The mass fraction of Xe in **B** is approximately 60%. The cation of **B** is obtained from the cation of **A** by reacting with the electrically neutral species **C**, a process involving a single proton transfer.

Which of the following statements is correct:

- A.** The cation of **A** is linear in shape.
- B.** The relative molecular mass of **A** is smaller than that of **C**.
- C.** The cations of **B** are triangular in shape.
- D.** The chemical reaction equation for the formation of **B** (with coefficients in the simplest whole-number ratio), where the sum of the coefficients on the right side of the equation is 3.
- E.** **C** contains 3 types of elements.
- F.** The number of atoms contained in **C** is 4

Answer

Correct Answer: **F**

Detailed Explanation

The formation of **A** is straightforward. Since it is a 1:1 reaction, XeF_2 can only insert into XeF^+ to yield Xe_2F_3^+ , which is the cation of **A**. According to VSEPR theory, one F atom bridges the two Xe atoms and adopts sp^3 hybridization, resulting in a trigonal shape rather than a linear geometry. Therefore, option A is incorrect.

CHECKPOINT

2 PTS

The cation of **A** is Xe_2F_3^+ with a trigonal shape

The anion of **B** is the same as that of **A**, both being hexafluoroarsenate anions. The cation contains oxygen with a coordination number of 2, suggesting it is composed of Xe, O, and F. Combining the mass fraction, the chemical formula of **B**'s cation is determined to be Xe_3OF_3^+ .

CHECKPOINT

2 PTS

The chemical formula of **B**'s cation is Xe_3OF_3^+

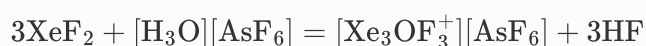
Considering its structure, oxygen cannot be a terminal atom and must act as a bridging atom. Based on the structure of Xe_2F_3^+ , Xe_3OF_3^+ can be derived by replacing a terminal fluorine atom in Xe_2F_3^+ with an oxygen atom and then attaching a XeF unit, described as $\text{F}-\text{Xe}-\text{F}-\text{Xe}-\text{O}-\text{Xe}-\text{F}$. According to hybridization theory, it adopts a bent shape, making option C incorrect.

CHECKPOINT

1 PTS

Xe_3OF_3^+ has a bent shape

Thus, the reaction for forming **B** can be written as:



The sum of the coefficients on the right side of the equation is 4, so option D is incorrect.

Xe_3OF_3^+ is generated from Xe_2F_3^+ and **C** via a proton transfer process. Given that Xe_3OF_3^+ is equivalent to replacing a terminal fluorine atom in Xe_2F_3^+ with an oxygen atom and attaching a XeF unit, the displaced fluorine atom likely combines with the proton. Therefore, the structure of **C** is clearly HOXeF , containing four elements and four atoms. Thus, option F is correct, while options E and C are incorrect.

CHECKPOINT

1 PTS

The structure of **C** is HOXeF